Data Analysis - Agentic Capital Allocation

1 Objective

Main idea that we want to analyze:

ERC-8004 Agents → Operating Wallets → x402/Financial Activity → Reputation and Information.

The primary tasks are to extend the ERC-8004 data, identify operating wallets, reconstruct financial and x402 activity, match the datasets, classify financial flows, and report the resulting sample coverage.

We don't need to extend the bot-versus-human classification, payment-rail comparison, or real-options simulation at this stage.

2 ERC-8004 Agent Data

Use the existing downloaded ERC-8004 dataset as the starting and validation sample. Extend the data through the latest complete date using:

- published extraction code: https://github.com/YulinLiu20/ERC8004;
- multi-chain event archive: https://github.com/qntx/erc8004; and
- official contracts and ABIs: https://github.com/erc-8004/erc-8004-contracts.

Begin with Ethereum and Base.
For each agent, construct:

- Agent ID, chain, registration date/block/transaction;
- time-varying owner wallet;
- operating wallet, where identifiable;
- metadata and declared services; and
- registration, ownership-transfer, and wallet-update events.

Do not automatically treat the owner wallet as the operating wallet. Record the source and confidence of each agent-wallet mapping.

Produce:

- `erc8004_agents`;
- `erc8004_events`; and
- a coverage table reporting agents, owners, identified operating wallets, metadata, services, reputation coverage, and ownership changes by chain.

3 Reputation and Feedback

Reconstruct the complete timestamped ERC-8004 feedback history.

For each feedback event, collect:

- Agent ID and feedback-provider wallet;
- timestamp, block number, and transaction hash;
- feedback value, decimals, and tags;
- service/endpoint information, where available; and
- revocation or modification status.

Construct an agent-day reputation file containing cumulative feedback count, average feedback within comparable categories, positive/negative feedback shares, unique providers, provider concentration, feedback dispersion, and time since most recent feedback.

Flag self-feedback, repeated feedback, unusually concentrated providers, revoked feedback, and obvious mechanical or spam-like patterns. Preserve both raw observations and flags.

Produce:

- `erc8004_feedback`; and
- `erc8004_reputation_daily`.

Report total feedback observations, agents with feedback, unique providers, major feedback categories, and the distribution of feedback across agents and providers.

4 Agent Wallets and Financial Activity

For ERC-8004 agents with reliably identified operating wallets, reconstruct the complete available blockchain transaction history. Validate the extraction on a small sample before extending it to all identified wallets.

Preserve transaction-level records containing:

- agent and wallet address;
- timestamp, block number, and transaction hash;
- sender and receiver;
- token and amount;
- transaction direction; and
- relevant contract interactions.

Construct an agent-day panel containing total inflows and outflows, stablecoin inflows and outflows, wallet balance, transaction count, unique counterparties, and active days.

Produce:

- `agent_transactions`; and
- `agent_wallet_daily`.

5 x402 Payment Data

Extract transaction-level x402 payment data, initially focusing on Base mainnet and USDC. Use the full available history through the latest complete date.

Begin with the following sources:

- Official x402 protocol and documentation:
  https://github.com/x402-foundation/x402
- Public Base transaction extraction code:
  https://github.com/MarkovianProtocol/x402-second-measurement
- Additional settlement and service data:
  https://github.com/RegardV/x402-recorder
- Aggregate data for validation:
  https://x402stats.io/data

Construct one observation per x402 payment containing, where available:

- timestamp, block number, and transaction hash;
- payer and payee wallet addresses;
- payment amount and token;
- chain;
- facilitator; and
- service or resource information.

Document the rule used to identify x402 payments and validate a random sample of transactions against blockchain records.

Match to ERC-8004

Match x402 payer and payee addresses to the operating wallets identified in the ERC-8004 data using exact wallet-address matches. Do not use owner wallets to assign transactions to individual agents unless the mapping to a specific agent can be independently established.

Report:

- total x402 transactions, payment volume, payers, and payees;
- number of ERC-8004 agents matched to x402;
- number matched as payers, payees, and both;
- number of matched agents with repeated transactions; and
- number of matched agents with both x402 activity and reputation/feedback histories.

Preserve both the complete x402 transaction dataset and the matched ERC-8004–x402 dataset.

6 Classifying Financial Flows

Classify inflows to identified agent wallets into:

- owner funding;
- customer/x402 revenue;
- payments from other identified agents;
- internal or related-wallet transfers;
- refunds;
- other identified transfers; and
- unclassified inflows.

Use ERC-8004 ownership information, x402 transactions, counterparties, transaction direction, and contract interactions.

For each transaction, record the flow classification, classification method, and confidence. Do not classify uncertain inflows as funding or revenue.

Report the number and value of flows in each category and the percentage of total flows that can be classified with reasonable confidence.

7 Data Validation and Sample Construction

Manually validate a random sample of at least 20 ERC-8004 agents and 50 x402 transactions against blockchain records or another independent source.

Verify:

- agent ID, registration date, and ownership;
- agent/operating wallet identification;
- ownership or wallet changes over time;
- reputation/feedback events and timestamps;
- wallet and x402 transactions;
- token amounts and decimals;
- duplicate agents, transactions, or feedback events; and
- obvious spam, Sybil, or mechanically generated activity.

Document all material discrepancies and any corrections made to the extraction procedure.

Sample Construction

Report the number of agents remaining at each relevant stage:

Table 1: Preliminary Sample Construction

Sample Criterion	Number of Agents
All ERC-8004 agents	
Agents with identified operating wallet	
Agents with financial activity	
Agents with reputation/feedback	
Agents with x402 activity	
Agents with both feedback and financial activity	
Agents with both feedback and x402 activity	
Agents with at least 30 days of observable history	

Do not impose additional sample restrictions at this stage.

Also report basic distributions of agent activity, including transactions, payment volume, feedback observations, counterparties, and active days. Report the 10th, 25th, 50th, 75th, 90th, and 99th percentiles.

8 Final Data Report

Submit the cleaned datasets together with a concise report containing:

- ERC-8004 agents and unique owners by chain;
- agents with reliably identified operating wallets;
- agents with financial activity;
- agents with timestamped reputation/feedback;
- total x402 transactions, volume, payers, and payees;
- ERC-8004 agents matched to x402;
- agents with both reputation and x402/financial activity;
- percentage of financial flows classified as funding, revenue, agent-to-agent payments, or other;
- agents with 7, 14, and 30 days of observable history; and
- agents for which a complete agent-day panel can be constructed.

Report the sample funnel from registered ERC-8004 agents to the final matched longitudinal sample and identify the major sources of sample loss.

Conclude with the major data limitations and a short list of empirical tests that the resulting data could support. Do not estimate the main regressions at this stage.